Central charge windows in the memory spectrum
A uniform trace bridge and the two force orientations

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Abstract

Balanced occupation constraints transfer each fixed marked Wick graph from the full-Fock trace to a compact central charge window with an $O(p^{-1})$ absolute error in its normalized graph weight. An exact general-charge Casimir identity then bridges the output deficit to the two physical force orientations. Existing marked-chord kernels therefore determine the central oriented spectral matrix and the complete two-energy measure of the projected second-chaos memory core. The raw second memory coefficient still requires restoration of its other descendants at central charge.

1 State and the dynamical result

Use standard CAR, the full overlapping complex Gaussian p-body Hamiltonian, common couplings in all sectors, and $\hbar = 1$. Let N be even, $\lambda_N = p^2/N \to \lambda > 0$, $m = N/2 + xp$, $|x| \le L_0$, and $C_m = \mathbb{E}\,\mathrm{tr}_m\, H_m^2 \to C = J_s^2/\lambda$. The trace is normalized maximally mixed at charge m; all norms include the Gaussian average. Constants below are uniform for x in this window and λ_N in positive compact sets.

Define $d_i = [H, c_i]$ before restricting neutral operators, and

$$A_- = \sum_i d_i^\dagger d_i, \quad A_+ = \sum_i d_i d_i^\dagger, \quad O_\pm = \mathsf{W}_2 A_\pm, \quad B = \mathsf{W}_2 h^2, \quad h = H_m. \tag{1}$$

Here W_2 is ordinary coupling-coordinate second chaos. Put $q = e^{-4\lambda}$, $c = 2\lambda(1-q)/(1+q)$ and $Y_\pm = O_\pm - cB$. The central two-energy limit is

$$\boxed{d\nu_{(Y_-,Y_+)} = \begin{pmatrix} w & \eta \\ \eta & w \end{pmatrix} d\mu_{q^2}, \qquad \begin{aligned} w &= 4C^2\lambda + \frac{16C^2\lambda^2 q}{1+q}, \\ \eta &= -4C^2\lambda q + \frac{16C^2\lambda^2 q}{1+q}. \end{aligned}} \tag{2}$$

The normalized Mehler measure μ_r is the existing probe measure with Hamiltonian variance C and chord parameter q. Its support in each energy is $[-2\sqrt{C/(1-q)}, 2\sqrt{C/(1-q)}]$. It is the prior chord kernel, now applied to the present state. For the output-deficit residual T defined on page 5,

$$d\nu_T = \frac{C^2 v \Delta}{v + \Delta}\, d\mu_{q^2}, \quad v = 2\lambda(1-q) + \frac{16\lambda^2 q}{1+q}, \quad \Delta = 2\lambda(1+q). \tag{3}$$

Both this variance and the eigenvalues of (2) are positive.

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2 The occupation ratio on a central window

For a term fixing t occupied and $L-t$ empty sites, the exact sector/full-Fock trace ratio is

$$R_{N,m}(L,t) = 2^L \frac{(m)_t^{\downarrow}(N-m)_{L-t}^{\downarrow}}{(N)_L^{\downarrow}} = 2^L \frac{\binom{N-L}{m-t}}{\binom{N}{m}}. \tag{4}$$

Let $s = L-t$, $d = 2t-L$ and $\epsilon = (m-N/2)/N$. For $L \le kp$ at fixed k, all relevant falling factorials are positive for sufficiently large p. An explicit first approximation is

$$\begin{aligned} A = {} & t\log(2m/N) + s\log(2(N-m)/N) \\ & - \frac{t(t-1)}{2m} - \frac{s(s-1)}{2(N-m)} + \frac{L(L-1)}{2N}. \end{aligned} \tag{5}$$

The elementary log remainder gives

$$|\log R - A| \le E(m,t) + E(N-m,s) + E(N,L), \quad E(a,j) = \frac{j(j-1)(2j-1)}{12a^2[1-(j-1)/a]}, \tag{6}$$

with $E(a,0) = 0$. This follows by summing $|\log(1-y)+y| \le y^2/[2(1-y)]$ for the three factorials. The denominator alone also gives the global upper bound

$$\log R \le L\log(1+2|\epsilon|) + \frac{L(L-1)}{2(N-L+1)} \le K_{k,L_0,\lambda}. \tag{7}$$

Expanding only the inverse denominators in A yields

$$\begin{aligned} \log R = {} & \frac{L}{2}\log(1-4\epsilon^2) + \frac{d}{2}\log\frac{1+2\epsilon}{1-2\epsilon} + \frac{L-d^2}{2N} \\ & + O\left(|\epsilon|L^2/N + L^3/N^2\right). \end{aligned} \tag{8}$$

Since $|\epsilon| \le L_0 p/N$, this proves, uniformly in $0 \le t \le L$,

$$|\log R| \le K\left[\frac{1+|d|}{p} + \frac{d^2}{N}\right], \quad |R-1| \le e^K |\log R|. \tag{9}$$

The last inequality uses the bounded positive part of $\log R$; no lower bound on R is needed. A balance estimate on d will turn this into an integrable error for the actual marked graphs.

3 Transferring fixed marked energy moments

Fix a Gaussian pairing and a finite number of normal or reverse commutator marks. Each Gaussian chord carries two p-element sets of opposite orientation. Let K_{ov} be the sum of pairwise intersection sizes of the distinct sets in this fixed graph. For a nonzero local occupation term, singleton sites on the two sets of a chord have opposite occupations and equal original cardinalities. Removing the multiply used sites changes this balance by at most a graph-dependent multiple of K_{ov}. Thus

$$|2t - L| \le c_k K_{\mathrm{ov}}, \qquad L \le kp. \tag{10}$$

For the Wick blocks used here there is no Gaussian chord internal to a marked block. A force mark, or a fixed number of body-deficit marks, is supported on intersections and annihilates singleton sites. Expanding multiply used sites into occupation projectors has absolute coefficient bound $(1 + K_{\mathrm{ov}})^j e^{c_k K_{\mathrm{ov}}}$ relative to the common singleton normalization. Fermionic signs and the orientations are retained; this is an absolute error estimate on their sum.

The previously used exponential overlap bound can be seen from the hypergeometric factorial moments of a pair overlap K_{ij}:

$$\mathbb{E}(K_{ij})_r^{\downarrow} = \frac{(p)_r^{\downarrow,2}}{(N)_r^{\downarrow}} \le \left(\frac{p^2}{N-p}\right)^r. \tag{11}$$

Its factorial-moment generating function bounds $\mathbb{E}e^{aK_{ij}}$ uniformly for each fixed a; Hölder handles the finite sum of pair overlaps. Combining (7), (9) and (10) consequently gives

$$|\mathcal{G}_{m,N} - \mathcal{G}_{\mathrm{Fock},N}| \le \frac{K}{p}\mathbb{E}\left[(1 + K_{\mathrm{ov}})^{j+2} e^{c_k K_{\mathrm{ov}}}\right] \le \frac{K'}{p}. \tag{12}$$

Here $\mathcal{G}$ denotes the graph weight with its Gaussian couplings and singleton powers normalized. Apply this also to H^2 to see that the sector and full-Fock variance normalizations agree up to $O(p^{-1})$. This does not require summing over all charge sectors or assuming a tail estimate outside the window.

The oriented-word and Poisson-overlap rules are prior results [1]; their marked versions and endpoint certificates are already recorded in this project [2]. Equation (12) transfers those results to the central trace for every fixed left/right energy degree. It applies to bounded coefficient combinations of fixed marked Wick blocks, including $O_{\pm}$, B and the previously defined fourth-chaos Q_4. It does not automatically control a raw counterterm multiplied by a growing factor of p, nor a number of energy insertions growing with N.

4 The output deficit and the actual reverse force

For neutral full-Fock operators define

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \overline{D}X = (N-Q)X - \sum_i c_i X c_i^\dagger. \tag{13}$$

At arbitrary charge m, CAR gives the exact identities

$$\mathcal{C}_x X = (N-m+1)DX + m\overline{D}X - D\overline{D}X, \quad DH = pH, \quad (\overline{D}h)_a = \frac{x_a - p(N-m)}{m-p+1} h_a. \tag{14}$$

The compositions are evaluated before restriction. For $u = p - a$ put

$$D_{\mathrm{in}} h = \sum_a u h_a, \quad E_{\mathrm{in}} = \sum_a \frac{u(u-1+N-2m)}{m-p+1} h_a. \qquad (p - \overline{D})h = 2D_{\mathrm{in}}h + E_{\mathrm{in}}. \tag{15}$$

Unlike exact half filling, E_{in} is now $O_{L^r}(p^{-1})$ for every fixed trace/Gaussian moment order needed here. The factorial input-rank tail proves this from (15); the half-filled $O(N^{-1})$ bound must not be substituted.

Let $\mathcal{S} = \sum (a+b-r)\mathsf{W}_2 Z^+_{ab;r}$ and let $\widetilde{\mathcal{S}}$ use $(x_{a+b} - x_r)/N$ instead. Define, before restriction,

$$\overline{A} = (2p - \overline{D})H^2 = A_+ + \{H, (p-\overline{D})H\}, \tag{16}$$

$$E_{\mathrm{dbl}} = (2p - D)\overline{A}, \qquad U_{\mathrm{in},2} = \{h, D_{\mathrm{in}}^2 h\} + 2(D_{\mathrm{in}}h)^2. \tag{17}$$

Substituting into (14) gives the general-charge bridge

$$\begin{aligned} N\left(\widetilde{\mathcal{S}} - \frac{O_- + O_+}{2}\right) = {} & (N/2 - m + 1 - 2p)O_- + (m - N/2 - 2p)O_+ \\ & + (2m - N - 1)\mathsf{W}_2\{h, D_{\mathrm{in}}h\} - \mathsf{W}_2 U_{\mathrm{in},2} + \mathsf{W}_2 E_{\mathrm{dbl}} \\ & + (m - 2p)\mathsf{W}_2\{h, E_{\mathrm{in}}\}. \end{aligned} \tag{18}$$

The double mark is supported on overlap sites: each deficit annihilates a singleton and a preceding deficit does not create a new singleton contribution. Its coefficient bound is $K(1+K_{\mathrm{ov}})^2$, so (12) bounds its fixed moments. All other terms are controlled by the established input tails and Gaussian product bounds. Divide (18) by N and use $m - N/2 = O(p)$. The positive output moments also give $\|\mathcal{S} - \widetilde{\mathcal{S}}\|_2 = O(p^{-1})$. Hence

$$\boxed{\mathcal{S} = (O_- + O_+)/2 + O_{L^2}(p^{-1})} \tag{19}$$

uniformly on the central window, with bounded physical C_m.

5 The spectral measure of the projected quadratic core

The transferred force graph calculation gives (2) for all fixed mixed energy moments. The existing Q_4 endpoint certificates also give $\nu_{Q_4,Y_-} = \nu_{Q_4,Y_+} = 0$ centrally. These are entire cross measures: finite-system matrix measures

$$\nu_{ij,N}(dE, dE') = \mathbb{E}\,\mathrm{tr}_m[X_i \Pi_h(dE) X_j \Pi_h(dE')] \tag{20}$$

are positive. Uniform diagonal moments give tightness and uniform integrability, and the candidate compactly supported measures are moment determinate. Spectral Cauchy–Schwarz controls the off-diagonal measures. A subsequence argument gives convergence uniformly over the stated central parameters.

Let Π_N be the actual finite-N orthogonal projector onto $\mathrm{span}(B, L_{\mathrm{in}}, O_-)$ and $T_N = (1-\Pi_N)\mathcal{S}$. The previous complete central Gram has a nondegenerate limiting inverse. With $\rho = v/(v+\Delta)$,

$$T_N = \left(\frac{1}{2} - \rho\right) Y_- + \frac{1}{2} Y_+ + o_{L^2}(1). \tag{21}$$

The L_{in} projection coefficient tends to zero. Multiplying the 2×2 matrix in (2) proves (3) and

$$\nu_{T,Y_-} = 0, \qquad \nu_{T,Q_4} = 0. \tag{22}$$

For an L^2 replacement the difference of self measures is bounded in total variation by $\|X - X'\|_2(\|X\|_2 + \|X'\|_2)$. Thus (19) and the projection replacement transfer the weak spectral limit and all bounded spectral tests, without expanding a time exponential. Unbounded polynomial tests on a replaced field require their own uniform integrability; the marked fields have it from (12).

Recall the actual raw core

$$\mathcal{E} = \sum_i [H, [H, c_i]]^\dagger [H, [H, c_i]] + [h, [h, A_-]]/4. \tag{23}$$

The established central strong expansion implies $\mathcal{R}_{2,N} := (1 - \Pi_N)\mathsf{W}_2\mathcal{E} = 8\lambda_N C_m q_N^2 T_N + o_{L^2}(1)$. Therefore

$$\boxed{d\nu_{\mathcal{R}_2} = V_2\, d\mu_{q^2}, \quad V_2 = 64\lambda^2 C^4 q^4 \frac{v\Delta}{v+\Delta}, \quad C_{\mathcal{R}_2}(t) = V_2 G_{q^2}(t).} \tag{24}$$

The limiting measure has no static atom at fixed $0 < q < 1$. Its normalized second frequency moment is $2C(1-q^2)$. With $C = J_s^2/\lambda$, after DS and then $\lambda \downarrow 0$, $V_2 \to 1024 J_s^8$ and $G_{q^2}(t) \to \mathrm{sech}^4(2J_s t)$. Equation (24) identifies the projected second-chaos core; it is not yet a restoration of the complete central raw U_2.

6 Executed checks and continuation

Three new scientific records are stored in `results/susskind-memory-central-spectrum`:

- `trace-ratio.json`: 24 exact finite binomial ratios and 324 central cases, with $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, $x = -2, 0, 2$, $L/p = 2, 4, 8$, and imbalance $2t - L = 0, 2, 6$. Every case satisfies the explicit log remainder (6) and upper bound (7). The 90/130-digit refinement differs by at most $2.23\,10^{-85}$.
- `force-bridge.json`: 108 exact cleared-rational identities on six non-half-filled (N, p, m) systems and three integer Hermitian inputs each. The checks use full CAR and complete Casimir projectors, including the general E_{in} correction and the pre-restriction compositions in (14).
- `spectrum-assembly.json`: symbolic oriented matrix, residual projection, zero cross with Y_-, and core normalization. It verifies the hashes of the prior local-mark and endpoint certificates rather than rerunning them. The reverse certificate retains 128 templates and 6,072 words; its coefficients vanish for arbitrary fixed energy degrees by the original endpoint proof.

For $\lambda = 0.5$, $x = 2$, $L = 4p$ and $2t - L = 2$, the actual trace ratios at $p = 16, 32, 64, 128$ are approximately 0.7985081, 0.9027791, 0.9522632, 0.9763479. The corresponding $p(R - 1)$ values approach a finite number near -3 in this example. Finite-size differences are retained. The analytic remainder bounds are not interval-rounding certificates; the numerical evaluations use precision refinement.

The next missing pieces for the raw central U_2 are its lower-chaos descendant table and exact conserved subtraction, and the filtered linear-field interfaces needed in its mixed spectral matrix. The new trace theorem handles fixed marked blocks but must not silently be multiplied by a divergent counterterm. These tasks precede the charge tails, finite-memory two-insertion remainder, and Susskind's bulk meson space/mass reconstruction. No bound uniform in growing word length, growing $|x|$, or joint small λ and finite N is asserted.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sec. 3.3, Eqs. (3.11)–(3.15). Existing oriented trace and overlap rules.

[2] A. Day and Chengheng, local project notes *Two orientations of the fermionic force*, *Central charge windows*, and *Closing the central-window memory Gram* (6 November 2026). Their formulas and executed certificates are inputs to this extension.